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The Influence of Institutional Investors on Myopic R&D Investment Behavior

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ABSTRACT: This paper examines whether institutional investors create or reduce incentives for corporate managers to reduce investment in research and development (R&D) to meet short-term earnings goals. Many critics argue that the frequent trading and short-term focus of institutional investors encourages managers to engage in such myopic investment behavior. Others argue that the large stockholdings and sophistication of institutions allow managers to focus on long-term value rather than on short-term earnings. I examine these competing views by testing whether institutional ownership affects R&D spending for firms that could reverse a decline in earnings with a reduction in R&D. The results indicate that managers are less likely to cut R&D to reverse an earnings decline when institutional ownership is high, implying that institutions are sophisticated investors who typically serve a monitoring role in reducing pressures for myopic behavior. However, I find that a large proportion of ownership by institutions that have high portfolio turnover and engage in momentum trading significantly increases the probability that managers reduce R&D to reverse an earnings decline. These results indicate that high turnover and momentum trading by institutional investors encourages myopic investment behavior when such institutional investors have extremely high levels of ownership in a firm; otherwise, institutional ownership serves to reduce pressures on managers for myopic investment behavior.

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I. INTRODUCTION

Over the past decade, many critics have expressed concerns that myopic investment behavior by U.S. corporate managers has caused U.S. firms to fall behind German and Japanese firms in terms of competitiveness and technological development (Jacobs 1991, 7–8; Porter 1992, 5–6; Laverty 1996, 825). Myopic investment behavior (or "managerial myopia") refers to underinvestment in long-term, intangible projects such as research and development, advertising, and employee training for the purposes of meeting short-term goals (Porter 1992, 3–5). In this paper, I examine managerial myopia with respect to research and development (R&D) investment because lagging U.S. investments in R&D have triggered much of the concern over managerial myopia and because prior research provides some evidence that managers use R&D to meet earnings goals (Jacobs 1991, 3; Dechow and Sloan 1991). U.S. accounting rules require that R&D expenditures be immediately and fully expensed. Critics argue that this accounting treatment creates the managerial myopia problem because the U.S. capital markets have a short-term focus that creates pressure on managers to sacrifice R&D to maintain short-term earnings growth (Drucker 1986; Jacobs 1991, 36; Porter 1992, 24).

A widely cited source of this pressure is the frequent trading and fragmented ownership of institutional investors.[1] By acting as "traders" rather than "owners," institutional investors allegedly place excessive focus on short-term developments, leading managers to fear that an earnings disappointment will trigger large-scale institutional investor selling and result in a temporary undervaluation of the firm's stock price (Graves and Waddock 1990, 76–77; Jacobs 1991, 37–38; Porter 1992, 43–46). However, this view of institutional investor behavior is not universal. Others argue that the large stockholdings and sophistication of institutional investors allow them to monitor and discipline managers, ensuring that managers choose investment levels to maximize long-run value rather than to meet short-term earnings goals (Monks and Minow 1995, chap. 2; Dobrzynski 1993). This paper tests these two competing views by examining whether institutional investors create or reduce pressures on managers to reduce investment in R&D to meet short-term earnings goals.

Myopic investment behavior is a type of earnings management that is most likely to happen when managers face a trade-off between meeting earnings targets and maintaining R&D investment. To capture this context, I select a sample of firms with pre-R&D earnings that are below the prior year's level, but by an amount that could be reversed by reducing R&D. For these firms, I test whether the level of institutional ownership affects the likelihood that managers cut R&D to reverse the expected earnings decline, controlling for other motivations to cut R&D (e.g., changes in the firm's set of profitable R&D opportunities). The results indicate that managers are more likely to reduce R&D expenses, and hence boost earnings toward last year's level, when institutional ownership is low. This result

[1] As defined by the SEC in Rule 13-f, institutional investors are entities such as bank trusts, insurance companies, mutual funds, and pension funds that invest funds on the behalf of others and manage at least $100 million in equity. Entities such as arbitrageurs, brokerage houses, and companies holding stock for their own portfolio (as opposed to in their pension funds) are not considered institutional investors by the SEC and are not required to disclose their equity investments under Rule 13-f (Wines 1990).

suggests that institutional investors are sophisticated investors who typically fulfill a monitoring role in reducing incentives to manage earnings with cuts in R&D.

The absence of evidence that institutional ownership, as a whole, encourages myopic investment behavior does not preclude the possibility that predominant ownership by certain groups of institutions could increase incentives for managerial myopia. I classify institutions into groups based on their past investment behavior to test whether institutions with short (long) investment horizons encourage (discourage) myopic investment behavior. The results indicate that predominant ownership by "transient" institutions—which have high portfolio turnover and engage in momentum trading strategies (i.e., buy (sell) firms with good (bad) earnings news)—significantly increases the likelihood that managers cut R&D to manage earnings. This result supports the assertion that heavy institutional trading based on current earnings leads to myopic investment behavior by corporate managers, but only when institutions exhibiting such behavior have concentrated ownership positions.

These findings contribute to the literature on the relation between a firm's ownership and its investment in R&D. Prior research in this area tests for systematic underinvestment in R&D by examining the cross-sectional relation between the level of R&D intensity (e.g., R&D-to-sales) and the level of institutional holdings. The evidence on this relation is mixed, though largely supporting a positive association between R&D intensity and the percentage of institutional holdings (Graves 1988; Baysinger et al. 1991; Hansen and Hill 1991; Wahal and McConnell 1997). I extend this line of inquiry by testing for period-specific underinvestment in R&D in years when managers face a trade-off between maintaining R&D and meeting earnings targets. In doing so, I present evidence that the relation between institutional ownership and changes in R&D differs depending on the firm's current earnings performance.

This paper also contributes to recent research focusing on the influence of institutional investors on earnings management. Rajgopal and Venkatachalam (1997) present evidence consistent with greater institutional ownership reducing the incidence of a lower-cost form of earnings management: discretionary accruals. I focus on a more costly means of earnings management, R&D cuts, which have real implications for long-term value and, therefore, are of greater concern to equity holders like institutional investors. In a sample of the top 100 R&D spenders among NYSE/AMEX manufacturing firms, Bange and De Bondt (1997) find that greater institutional ownership is negatively associated with unexpected changes in R&D that reduce the anticipated gap between analysts' earnings forecasts and reported net income. I provide evidence that this relation holds for a broader set of firms and that the relation specifically holds for firms that reduce R&D to increase earnings. I also extend this research by providing evidence that ownership by different groups of institutional investors has differential influences on the use of R&D to manage earnings.

Finally, this paper makes two methodological contributions. First, prior research examining the use of R&D to manage earnings generally relies on simple proxies, such as industry changes in R&D, for nondiscretionary changes in R&D (Dechow and Sloan 1991; Holthausen et al. 1995). I use prior work on the determinants of R&D spending to develop a more extensive model for the firm's set of profitable R&D opportunities and the costs of reducing R&D to manage earnings, leaving an unexplained portion that is more likely to capture the changes in R&D resulting from earnings management. Second, this study introduces a methodology by which institutional investors can be parsimoniously classified into groups based on past portfolio behavior (e.g., trading frequency, level of diversification). By classifying directly on such characteristics, this approach creates more homogeneous groups of institutions and allows for richer predictions than the classification of institutions by type (e.g., pension funds, mutual funds) commonly used in prior research (Lang and McNichols 1997).

The next section develops hypotheses by examining arguments for how the behavior of institutional investors influences managerial incentives to engage in myopic R&D investment behavior. Section III presents the methodology used in the empirical tests. Section IV describes the sample and presents descriptive statistics. Section V reports results of the empirical tests for institutional ownership as a whole, and Section VI reports results for ownership by different groups of institutions. Section VII presents a summary and conclusions.

II. HYPOTHESIS DEVELOPMENT

Short-Term-Oriented Institutional Investor Behavior and Managerial Myopia

The first hypothesis tests the view that the short-term focus of institutional investor trading creates incentives for myopic investment behavior. This view argues that managers have incentives to reduce R&D in order to avoid an earnings disappointment that would trigger large-scale institutional investor selling and lead to a temporary misvaluation of the firm's stock price (Graves and Waddock 1990, 76–77; Jacobs 1991, 37–38; Porter 1992, 43–46). This hypothesis requires that (1) managers have incentives to avoid temporary misvaluations and (2) institutional investor trading is sensitive to current earnings news and can cause a temporary misvaluation.

Prior research identifies two conditions necessary to create incentives for myopic investment behavior to avoid temporary misvaluations (Stein 1988, 1989). First, managers must place enough weight on current stock price so that they would not be willing to "ride out" a temporary misvaluation. Such concern over current stock price could be driven by (1) stock-based compensation and the need to make liquidity trades, (2) near-term equity funding requirements, (3) the threat that a raider will exploit a temporary undervaluation, or (4) the time horizons of influential investors (Froot, Perold, and Stein 1992, 50–55). Second, managers must believe there is a potential for misvaluation based on current earnings, either through market overreactions to unmanaged earnings declines or through market underreactions to reductions in valuable R&D that artificially boost earnings. Support for this condition is provided by Bernard et al. (1993), who cite anecdotal evidence that managers believe the market is capable of overreacting to earnings news.

The presence of institutional investors increases the perceived potential for misvaluation if they trade heavily based on current earnings news and if such trading causes misvaluations. Prior research provides evidence supporting both of these conditions. Lang and McNichols (1997) find that institutional trading is sensitive to earnings news, even after controlling for return movements. Other studies report that increased institutional ownership is associated with higher trading volume and stock return volatility around quarterly earnings announcements, as well as temporary undervaluations after spin-offs (Kim et al. 1997; Potter 1992; Brown and Brooke 1993).[2] Such heavy institutional investor trading in response to reported earnings could result from several market frictions. First, institutions subject to strict fiduciary responsibilities have incentives to sell stocks with declining earnings because fund sponsors and the courts use earnings as an objective criterion to judge the prudence of an investment (Badrinath et al. 1989). Second, institutional investors sometimes use

[2] Eames (1997) reports that annual changes in institutional ownership do not affect earnings response coefficients (ERCs) in a manner consistent with institutional trading leading to earnings-based misvaluations. However, in using annual data, his study does not capture the effect of institutional trading on ERCs in the days or weeks surrounding earnings announcements, when trading-based misvaluations are most likely to occur.

current earnings as a "value proxy" in their trading decisions due to an information asymmetry surrounding the quality of R&D spending (Froot, Perold, and Stein 1992, 55–56; Porter 1992, 43). This asymmetry could arise if institutions have short expected holding periods and focus on predicting near-term price movements instead of valuing long-term prospects (Froot, Scharfstein, and Stein 1992; Porter 1992, 43). Also, if institutions hold diversified portfolios with numerous stocks and small stakes in each, research on the long-term prospects of each is less cost-effective (Froot, Perold, and Stein 1992, 56; Porter 1992, 43).

These arguments for the extreme sensitivity of institutional investor trading to current earnings news support the hypothesis that, relative to individual investors, institutional investors create a short-term-oriented environment that encourages managers to manipulate earnings through investment cuts:[3]

> **H1a:** *Ceteris paribus,* a higher percentage of institutional holdings in a firm *increases* the likelihood that its manager cuts R&D to meet short-term earnings goals.

Sophisticated Institutional Investor Behavior and Managerial Myopia

The competing hypothesis tests the view that the sophistication and large stockholdings of institutional investors remove incentives for myopic investment behavior by providing a higher degree of monitoring of managerial behavior. This monitoring can occur either explicitly, through governance activities, or implicitly, through information gathering and correctly pricing the impact of managerial decisions. Institutions that invest in firms with the intention of holding substantial ownership blocks over a long horizon have strong incentives to incur the cost of explicitly monitoring managers and ensure the firm does not cut profitable R&D to meet short-term earnings goals (Dobrzynski 1993; Monks and Minow 1995, chap. 2). The positive impact of such institutional monitoring is documented by Opler and Sokobin (1997), who present evidence that poorly performing firms targeted by the Council of Institutional Investors for shareholder activism have substantially improved profitability, greater asset divestitures, and reduced acquisition activity in subsequent years.

Sophisticated institutional investors can implicitly monitor managerial investment behavior by gathering information concerning the quality of R&D, thereby reducing the opportunities of managers to prevent a negative market reaction to an expected earnings decline through cuts in R&D. Hand (1990) uses institutional ownership to proxy for the type of sophisticated investor that would not be "functionally fixated" on earnings (i.e., focusing only on reported earnings and being misled by firms' accounting choices). Schipper (1989, 98) argues that:

> Concentrated user groups with substantial financial sophistication, material sums at stake, and no contractual friction to inhibit their behavior are, for example, likely candidates for undoing earnings management.

Considering the size of their investments and their use of buy-side analysts, institutional investors could fit this example and serve to remove incentives for myopic investment behavior.

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[3] Following Lang and McNichols (1997, 31), I assume that the other major category of investors in firms is individual investors. Other possible ownership groups include insiders, arbitrageurs and corporations making strategic investments. Agrawal and Knoeber (1996) find that the level of institutional holdings in a firm is not significantly associated with the level of insider holdings and is weakly negatively associated with large (greater than 5 percent) blockholders (which would include strategic corporate investments). Thus, institutional ownership likely does not proxy for the level of ownership by these types of investors.

This view of the sophistication of institutional ownership leads to the competing hypothesis that, relative to individual investors, institutional investors serve a monitoring role that reduces managerial incentives to sacrifice R&D to meet current earnings goals:

H1b: *Ceteris paribus,* a higher percentage of institutional holdings in a firm *decreases* the likelihood that its manager cuts R&D to meet short-term earnings goals.

The Influence of Different Groups of Institutional Investors on Managerial Myopia

Hypotheses 1a and 1b concern the impact of aggregate institutional ownership. Because institutional investors differ in their behavior and incentives, I examine whether certain groups of institutions influence myopic investment behavior in a manner different from the finding for overall institutional ownership. This examination is motivated by the growing importance of investor relation activities within firms and the increase in the number of consulting firms offering institutional investor targeting services. These trends indicate that managers not only know their investor base, but perceive differences in the ownership behavior of institutions and sometimes actively target certain types of institutions as investors (Anand 1991; Elgin 1992).[4] Lang and McNichols (1997) group institutions based on their type (e.g., investment advisers, bank trusts, pension funds) and find significant differences in portfolio turnover and earnings-based trading. To account for the possibility that there can be substantive differences in trading and governance behavior within types of institutions, I classify institutional investors into groups using the specific characteristics of institutional investor behavior that have been argued to increase the pressure on managers to manipulate earnings. This approach classifies institutions into three groups—transient, dedicated and quasi-indexer—based on their past investment patterns in the areas of portfolio turnover, diversification, and momentum trading.[5]

Porter (1992, 42–46) argues that pressures for myopic investment behavior are created by "transient" institutional owners, who hold small stakes in numerous firms and trade frequently in and out of stocks, generally basing their trades on a value proxy such as current earnings. The short-term focus of transient institutions makes them the most likely group to sell out of a firm with disappointing earnings news, creating incentives for managers to avoid earnings disappointments. This argument leads to the following hypothesis:

H2a: *Ceteris paribus,* if a firm's institutional ownership is predominantly composed of institutions that are transient, then its manager is *more* likely to cut R&D to meet short-term earnings goals.

In contrast, Porter (1992, 46–49) argues that "dedicated" owners (which he mainly ascribes to Germany and Japan) alleviate pressures for myopic investment behavior because their large, long-term holdings, which are concentrated in only a few firms, provide incentives to monitor managers and to rely on information other than earnings to assess the managers' performance. The monitoring role of dedicated institutions leads to the following hypothesis:

[4] For example, the consulting firm Georgeson Co. analyzes the portfolios of "almost 1,500 domestic and foreign institutions" in order to characterize a client's investor base and assist managers in "screening out" institutions with high turnover, index, or quantitative strategies in favor of institutions with long investment horizons and holdings in companies with similar characteristics to the client (Elgin 1992).

[5] This classification method does not proxy for classifications based on type, investment styles, or institutional size. Institutions in the transient, quasi-indexer and dedicated groups all exhibit significant heterogeneity across these other possible classification schemes. Because of this heterogeneity, I find no significant differences in results of the myopic behavior tests across groups of institutions formed by these other methods.

H2b: *Ceteris paribus,* if a firm's institutional ownership is predominantly composed of institutions that are dedicated, then its manager is *less* likely to cut R&D to meet short-term earnings goals.

The third group of institutional investors, "quasi-indexers," use indexing or buy-and-hold strategies that are characterized by high diversification and low portfolio turnover. Porter (1992, 42–46) claims that the presence of quasi-indexers exacerbates incentives for myopic investment behavior because their passive, fragmented ownership leads them to gather little information on the company and provides little incentive to monitor managers. This type of institutional ownership, in effect, abdicates its potential influence over managers to other types of active investors who could pressure managers into focusing on short-run earnings goals. However, Monks and Minow (1995, chap. 2) espouse the opposite viewpoint that institutions that are unable to sell because of their indexing strategies have strong incentives to monitor management to ensure it is acting in the long-term interests of the firm. Because of these contrasting views, I do not make a formal hypothesis about the influence of quasi-indexers, but include them in the analysis for descriptive purposes.

III. METHODOLOGY

Research Design

To test whether managers reduce R&D to meet earnings goals, I assume that managers use last year's earnings per share as their earnings target. The business press uses this benchmark to analyze current earnings performance and prior research provides evidence that managers have incentives to avoid decreases in earnings (Burgstahler and Dichev 1997; Barth et al. 1997).[6] I divide the sample into three subsamples based on the relationship between pre-tax, pre-R&D earnings changes and prior levels of R&D.[7] I test for myopic investment behavior in a sample of firms, which I call the "Small Decrease" (SD) sample, whose earnings before R&D and taxes have declined relative to the prior year, but by an amount that can be reversed by a reduction in R&D. This specification assumes that managers have unbiased expectations of pre-tax, pre-R&D earnings early enough in the fiscal year to make decisions to reduce R&D expenditures. Burgstahler and Dichev (1997) find that firms with small earnings decreases are underrepresented in the distribution of observed earnings changes, indicating that firms below but near last year's earnings are managing earnings upward. Baber et al. (1991) report that such firms increase R&D by less, on average, than firms with both increases or large decreases in earnings before investment, which they cite as evidence that managers of firms with small earnings declines use R&D cuts to increase earnings.

For comparison purposes, I also create "Increase" (IN) and "Large Decrease" (LD) samples. The IN sample contains all firms with positive changes in pre-tax, pre-R&D earnings; these firms could maintain last year's R&D and still have an increase in pre-tax earnings. The LD sample consists of firms with declines in pre-tax, pre-R&D earnings greater than the amount of prior year's R&D; these firms could eliminate R&D spending

[6] I chose not to use analysts' forecasts as the earnings target because this benchmark would (1) bias the sample toward firms with greater levels of institutional ownership (O'Brien and Bhushan 1990), (2) require an estimate of the analysts' expectation of R&D to get expected pre-R&D earnings, and (3) create a specification problem if analysts anticipate earnings management, as suggested by the findings of Burgstahler and Eames (1998).

[7] I use pre-tax earnings, rather than after-tax earnings, to make the results comparable to prior research (Baber et al. 1991) and to avoid introducing error in the estimation of marginal tax rates to obtain after-tax R&D. As a check, I estimated after-tax R&D using three tax rate assumptions: (1) the statutory rate, (2) the effective tax rate, and (3) the effective tax rate adjusted for the estimated R&D credit. The correlations between sample assignments across these three methods and the before-tax method were all over 0.85.

and still report a decrease in pre-tax earnings. Because managers of IN and LD firms do not have the same incentives to reduce R&D to meet the earnings target, the relation between institutional ownership and R&D cuts in these samples should differ from the relation in the SD sample if the relation is driven by institutional investor pressure for or against myopic investment behavior.

The dependent variable I use to test for myopic investment behavior is an indicator variable (CUTRD) that equals one if the firm cuts R&D relative to the prior year, and zero if the firm maintains or increases R&D. I use an indicator variable rather than the magnitude of the change in R&D because theory does not address how the level of institutional ownership should affect the magnitude of increases in R&D, and because the magnitude of decreases in R&D should be based on the amount of the earnings decline. Thus, the magnitude of the change in R&D is unlikely to be a linear function of the level of institutional ownership.[8]

To test hypotheses 1a and 1b on the influence of institutional ownership on myopic investment behavior, I use a logit model to regress CUTRD on the level of the firm's institutional ownership and a set of control variables (discussed below) that proxy for changes in the R&D investment opportunity set, differences in the costs of cutting R&D, and proxies for other earnings management incentives:

$$\begin{aligned}\mathrm{Prob}(\mathrm{CUTRD}_i = 1) = F(&\alpha + \beta_1\mathrm{PCRD}_i + \beta_2\mathrm{CIRD}_i + \beta_3\mathrm{CGDP}_i + \beta_4\mathrm{TOBQ}_i \\ &+ \beta_5\mathrm{CCAPX}_i + \beta_6\mathrm{CSALES}_i + \beta_7\mathrm{SIZE}_i + \beta_8\mathrm{DIST}_i + \beta_9\mathrm{LEV}_i \\ &+ \beta_{10}\mathrm{FCF}_i + \beta_{11}\mathrm{PIH}_i) \qquad (1)\end{aligned}$$

where $F(\cdot)$ is the logistic c.d.f.

CUTRD = 1 if R&D is cut relative to the prior year, 0 otherwise;
PCRD = prior year's change in R&D;
CIRD = change in industry R&D-to-sales ratio, defined using 4-digit SIC codes;
CGDP = change in gross domestic product;
TOBQ = Tobin's q (the sum of market value of common equity, book value of preferred equity, long-term debt, and short-term debt, divided by total assets);
CCAPX = change in capital expenditures;
CSALES = change in sales;
SIZE = market value of equity;
DIST = distance from earnings goal relative to prior year's R&D, defined as the change in pre-tax, pre-R&D earnings divided by prior year's R&D;
LEV = leverage (debt-to-assets);
FCF = free cash flow (cash flow from operations less capital expenditures, scaled by net tangible assets);
PIH = percentage of institutional holdings.

To test hypotheses 2a and 2b on the effect of different groups of institutions on myopic

[8] I also considered using an indicator variable for whether the firm successfully reversed the earnings decline with R&D cuts. I chose not to use this variable because I could not apply the model out of the SD sample as a robustness check. I reran the analysis with this indicator variable in the SD sample and the results for the institutional holdings variable were unchanged.

investment behavior, I first restrict the sample to firms with at least five percent institutional ownership to ensure that institutional ownership is likely to be influential in the firm. Then I add indicator variables for the composition of institutional ownership to equation (1). The indicator variable for each group (transient, quasi-indexer, dedicated) equals one when a firm's institutional ownership is dominated by this group, and zero otherwise. This specification more powerfully identifies firms in which a given group is likely to be influential (compared to the percentage ownership by each group) and avoids the problem that percentage ownership by the three groups is highly correlated. This approach yields the following regression:

$$\begin{aligned}\text{Prob}(\text{CUTRD}_i = 1) = F(\alpha &+ \beta_1\text{PCRD}_i + \beta_2\text{CIRD}_i + \beta_3\text{CGDP}_i + \beta_4\text{TOBQ}_i \\ &+ \beta_5\text{CCAPX}_i + \beta_6\text{CSALES}_i + \beta_7\text{SIZE}_i + \beta_8\text{DIST}_i + \beta_9\text{LEV}_i \\ &+ \beta_{10}\text{FCF}_i + \beta_{11}\text{PIH}_i + \beta_{12}\text{DQ5(Transient)}_i \\ &+ \beta_{13}\text{DQ5(Quasi-Indexer)}_i + \beta_{14}\text{DQ5(Dedicated)}_i) \qquad (2)\end{aligned}$$

where $\text{DQ5}(\text{GROUP}_k)_i = 1$ if firm i is in the top quintile of proportional ownership by group k, and zero otherwise; GROUP_k = Transient, Quasi-Indexer, Dedicated.

Control Variables

To control for motivations to cut R&D that are unrelated to myopic investment behavior incentives, I include proxies for the R&D investment opportunity set (i.e., the set of available positive NPV projects in which the firm could invest). These proxies are based on the expectations model for the level of R&D intensity developed in Berger (1993). Except for Tobin's q, I take first differences in his variables to obtain a model for the change in R&D. Each of the variables is defined to have a negative expected relation with the decision to cut R&D (see table 1 for variable definitions).[9]

The prior year's change in R&D (PCRD) proxies for changes in the firm's R&D opportunity set over time (Berger 1993). If a firm's R&D opportunity set has been declining (increasing) over time, a decline (increase) in R&D last year makes a firm more (less) likely to cut R&D this year.[10] The change in industry R&D intensity (CIRD) captures changes in the R&D opportunity set within the firm's industry and changes in the level of R&D spending needed to stay competitive within the industry (Berger 1993). Both explanations suggest that firms in industries with increasing (decreasing) R&D are expected to be less (more) likely to cut R&D. Changes in the Gross Domestic Product (CGDP) measure growth in the overall economy and proxy for increases in the level of technological progress in the economy (Berger 1993). In years when CGDP is high (low), firms are expected to have more (fewer) profitable R&D opportunities and thus be less (more) likely to reduce R&D. Berger (1993) includes Tobin's q (TOBQ) to capture the marginal benefit-to-cost ratio of undertaking new investment. As a proxy for the unobservable marginal TOBQ, Berger (1993) uses the average TOBQ, defined as the market value of the firm's equity and debt,

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[9] I drop two variables from his model: a measure of funds available for investment and an R&D tax credit usability variable. I drop the funds available variable because it is pre-R&D earnings plus depreciation, which is similar to what I use to partition the sample. I drop the credit usability variable because it does not explain cuts in R&D and it requires three years of data on deferred tax liability, thereby reducing the sample size. Reed and Plumlee (1998) find that tax motivations only lead to cuts in R&D when the firm's earnings are substantially above the prior year's level.

[10] An alternative explanation is that if the firm reduced R&D in the prior year, it becomes more costly to cut R&D again, indicating a positive relation between PCRD and CUTRD.

TABLE 1
Definitions of Variables

Variable	*Definition*	*Expected Sign*[a]
Indicator for cut in R&D (CUTRD)	1 if $(RD_t\text{–}RD_{t-1}) < 0$, 0 otherwise	
Prior change in R&D (PCRD)	$\ln(RD_{t-1})\text{–}\ln(RD_{t-2})$	–
Change in industry R&D intensity (CIRD)	$\ln(IRD_t/ISALES_t)\text{–}\ln(IRD_{t-1}/ISALES_{t-1})$	–
Change in capital expenditures (CCAPX)	$\ln(CAPX_t)\text{–}\ln(CAPX_{t-1})$	–
Change in sales (CSALES)	$\ln(SALES_t)\text{–}\ln(SALES_{t-1})$	–
Tobin's q (TOBQ)	$TOBQ_t$	–
Change in gross domestic product (CGDP)	$\ln(GDP_t)\text{–}\ln(GDP_{t-1})$	–
Firm size (SIZE)	$\ln(MVE_t)$	–
Distance from earnings goal (DIST)	$(EBTRD_t\text{–}EBTRD_{t-1})/RD_{t-1}$	+
Leverage (LEV)	$(LTD_t+STD_t)/ASSETS_t$	+
Free cash flow (FCF)	$(CFO_t\text{–Average } CAPX_{t-1 \text{ to } t-3})/CA_{t-1}$	–
Percentage of institutional holdings (PIH)	$ISHARES_t/TSHARES_t$	+/–
Indicator for predominant ownership by group k ($DQ5(GROUP_k)$), k = quasi-indexer, transient, dedicated	1 if $PIH(GROUP_k)/PIH$ in top quintile of distribution, 0 otherwise	+/–

RD_t = Research and development per share (46/54)[b]
IRD_t = Total research and development for all firms in same 4-digit SIC code, excluding the firm
$ISALES_t$ = Total sales for all firms in same 4-digit SIC code, excluding the firm
$CAPX_t$ = Capital expenditures per share (30/54)
$SALES_t$ = Sales per share (12/54)
$TOBQ_t$ = Tobin's q [(Market value of equity + preferred stock + total debt)/total assets] [(199*25 +130+9+34)/6]
GDP_t = Gross domestic product
MVE_t = Market value of equity (199*25)
$EBTRD_t$ = Earnings per share before taxes and R&D ((170+46)/54)
LTD_t = Long-term debt (34)
STD_t = Short-term debt (9)
$ASSETS_t$ = Total assets (6)
CFO_t = Cash from operations $(18-\Delta4+\Delta5+\Delta1-\Delta34+14)$
CA_t = Current assets (4)
$ISHARES_t$ = Total shares held by institutional investors
$TSHARES_t$ = Total shares outstanding
$PIH(GROUP_k)$ = Percentage institutional ownership by group k, k = Quasi-indexer, transient, dedicated

[a] *This column* lists each variable's expected relation to CUTRD.
[b] Compustat numbers are in parentheses.

divided by the book value of its assets. Because of the accounting treatment of R&D, this measure is similar to the market-to-book ratio in that both are positively associated with the future value of R&D spending. Based on these interpretations, firms with higher (lower) TOBQ have more (fewer) valuable R&D opportunities and face a higher (lower) cost of reducing R&D for myopic reasons.[11]

I include additional variables not found in Berger's (1993) model to control for other R&D opportunity set measures, for the costs of cutting R&D to manage earnings, and for other incentives to manage earnings. The change in capital expenditures (CCAPX) proxies for reduced funds available for investment and/or entry into a more mature, lower investment stage of the firm's life cycle (Perry and Grinaker 1994). Both explanations yield an expected negative relation with CUTRD. The change in sales (CSALES) is not included in Berger's (1993) model because he is estimating the R&D-to-sales ratio. I include CSALES to proxy for firm growth and funds available for R&D investment, as well as to capture the fact that R&D budgets are often based on sales (Berger 1993). For each of these reasons, a negative relation with the decision to cut R&D is expected.

A size variable (SIZE), the log of market value of equity, proxies for two possible effects, both of which lead to a negative expected relation with CUTRD. First, SIZE proxies for the amount of information available about the firm (Wiedman 1996). Larger (smaller) firms have richer (poorer) information environments that should reduce (increase) opportunities for successful earnings management with R&D. Second, SIZE proxies for the likelihood the firm faces cash constraints (Jalilvand and Harris 1984). Smaller firms are more likely to suffer cash flow shortages that force them to reduce R&D.[12]

The distance of the expected earnings from the earnings target (DIST) measures the percentage of R&D that would need to be cut to obtain a positive earnings change. As a firm's expected earnings get farther from (closer to) the earnings goal, the cost of cutting R&D to manage earnings increases (decreases). This variable is defined as the change in pre-tax, pre-R&D earnings divided by prior year's R&D. For firms in the SD sample, this variable ranges between 0 and -1. A firm with a value of DIST equal to -0.8 (-0.2) would need to reduce its R&D by 80 percent (20 percent) relative to the prior year to reverse the decline in earnings. Firms with high (less negative) values of DIST have the ability to meet their earnings target through less severe cuts in R&D (e.g., reductions in the number of prototypes or test runs and/or elimination of basic research projects); whereas firms with lower (more negative) values of DIST need to make more substantial cuts in

[11] I use the level of Tobin's q rather than the change in Tobin's q because the latter measure is highly correlated with changes in the market value of equity. Since such changes can be driven by factors other than changes in the R&D opportunity set (such as market reactions to the decision to cut R&D), the construct validity of the levels measure is likely to be higher. If I replace TOBQ with the change in Tobin's q, the coefficient on the latter is positive and significant, which is opposite of what is expected and likely reflects the construct validity problem. I also replaced TOBQ with the market-to-book ratio and found similar results. Neither of the changes affects the sign or significance of the institutional holdings variables.

[12] SIZE could also proxy for two effects that lead to a positive relation with CUTRD. First, smaller firms generally have larger managerial holdings, which reduces incentives to manage earnings based on capital market pressures (Warfield et al. 1995). Second, smaller firms tend to have higher (lower) percentages of new product research (basic research) in their R&D budgets (Mansfield 1981). To the extent that it is more (less) costly to cut R&D for new products (basic research), smaller firms have less flexibility to cut R&D to manage earnings (Schneiderman 1991). In both these cases, SIZE is a fairly weak proxy for the underlying effect, making a positive relation between SIZE and CUTRD less likely.

R&D (e.g., personnel and equipment) to meet their earnings goals. Thus, a positive relation between DIST and CUTRD is expected.[13]

The firm's leverage ratio (LEV) captures potential debt covenant incentives to manage earnings (Duke and Hunt 1990). A positive sign is expected if meeting debt covenants motivates earnings management. Higher LEV could also indicate fewer growth opportunities for the firm (Myers 1984), in which case a positive sign would also be expected. Finally, I include a measure of free cash flow (FCF), defined as cash flow from operations less capital expenditures, scaled by net tangible assets, to proxy for possible near-term financing requirements. Firms with substantially negative free cash flows have a greater need to raise equity in the near-term, and thus have incentives to boost earnings and reduce the chance of an undervaluation (Dechow et al. 1996). This variable also proxies for reduced funds available for investment, which might trigger reductions in R&D. Either explanation indicates that a negative relation with CUTRD is expected.[14]

Institutional Ownership Variables

I measure the percentage of institutional holdings (PIH) as total shares held by institutions divided by total shares outstanding. PIH is calculated at the end of the calendar quarter in which the firm's third fiscal quarter ends. This date is chosen to measure institutional ownership in the middle of the second half of the fiscal year, when I assume the manager likely has an accurate expectation of annual earnings and begins to consider myopic investment behavior decisions.[15]

To obtain variables on the percentage of ownership by different groups of institutions, I first use factor analysis and cluster analysis to assign institutions into groups based on their past investment behavior (see section VI for details of this procedure). Then, I calculate the proportion of ownership held by each group of institutions for each firm (e.g., $PIH(GROUP_k)_i/PIH_i$), where the groups are transient, dedicated, and quasi-indexer. I rank firms into quintiles based on the distribution of this proportion for each group and create an indicator variable ($DQ5(GROUP_k)_i$) that equals one if the firm is in the top quintile of proportional ownership by group k, and zero otherwise.[16]

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[13] The DIST variable captures other incentives outside of the SD sample. In the IN sample, DIST is always positive, and large values indicate a large positive change in pre-R&D earnings relative to prior R&D. If managers have incentives to increase R&D to dampen earnings growth, a positive sign should result. In the LD sample, DIST is always less than -1, and small values indicate a large negative change in pre-R&D earnings relative to prior R&D. If managers have incentives to increase R&D to take a "big bath" in earnings during the year, a negative relation should result.

[14] I chose not to include a proxy for bonus motivations because Holthausen et al. (1995) provide evidence that R&D is not managed for bonus reasons and because hand collection of a bonus plan proxy would significantly reduce the sample size.

[15] To obtain some descriptive evidence on this assumption, I examined quarterly R&D disclosures on Compustat. Compustat reports quarterly R&D for years after 1989, but quarterly R&D disclosures are voluntary and are only reported by approximately 40 percent of the sample firms. For firms in the SD sample, the mean seasonal percentage changes in R&D by quarter are 10.8 percent for the first quarter, 8.7 percent for the second, 7.3 percent for the third, and 4.8 percent for the fourth. For firms in the IN (LD) sample, the comparable changes are 12.1 percent, 12.8 percent, 14.6 percent and 14.8 percent (12.6 percent, 10.7 percent, 5.6 percent, and 7.0 percent), respectively. This evidence indicates that some SD sample firms are able to reduce R&D late in the fiscal year.

[16] I use quintiles because they yield extreme proportional differences while still maintaining a reasonable number of observations in the top quintile. I performed the ranking, and the subsequent analyses, using treciles and quartiles. The results are similar, though less significant, using the less extreme fractiles. I also formed the quintiles after ranking on SIZE and PIH to control for any association between extreme ownership by groups and those variables. The results were insensitive to this approach.

IV. SAMPLE AND DESCRIPTIVE STATISTICS

The sample includes all firm-years between 1983 and 1994 with available data. The sample period is restricted by the availability of institutional holdings data and the requirement of two years of data from which to calculate portfolio characteristics of institutions. I collect institutional holdings data from the University of Michigan Spectrum database, which contains all 13-f filings between September 30, 1981 and December 31, 1994. According to SEC Rule 13-f, all institutions managing more than $100 million in equity must file a quarterly report listing all equity holdings that are greater than 10,000 shares or $200,000 in market value. Thus, for each firm, total institutional holdings is defined as the sum of all end-of-calendar-quarter holdings by fund managers filing 13-fs on the firm. I gather total shares outstanding from the 1995 CRSP Daily NYSE, AMEX and NASDAQ files, and GDP data from the 1995 Economic Report of the President. All other variables are obtained from the 1995 Compustat PST, Full Coverage and Research files.

Panel A of table 2 shows the criteria used to select the sample. I exclude firm-year observations for which: (1) current or prior year R&D data is missing, (2) the ratio of R&D-to-sales is less than 1 percent (i.e., R&D is not a significant portion of earnings),

TABLE 2
Sample Sizes

Panel A: Sample Selection Criteria

Criteria	*Firm-Years*
Compustat firm-years with nonmissing R&D, 1983–1994	30,324
Less: Missing prior year R&D data	(3,754)
R&D/sales less than 1%	(6,135)
Fewer than three other firms in industry	(131)
Missing data to calculate independent variables	(4,165)
Missing Spectrum institutional holdings data	(2,195)
Final sample	13,944

Panel B: Number of Observations and Frequency of R&D Cuts in Small Decrease (SD), Increase (IN) and Large Decrease (LD) Samples[a]

	Samples			
	SD Sample	*IN Sample*	*LD Sample*	*Total*
Cut R&D[b]	852 (34.5%)	2,190 (25.7%)	1,041 (35.2%)	4083
Increase R&D	1,617 (65.5%)	6,330 (74.3%)	1,914 (64.8%)	9861
Total	2,469	8,520	2,955	13,944

[a] The SD sample consists of all firm-years for which $-RD_{t-1} < (EBTRD_t - EBTRD_{t-1}) < 0$. The IN sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) > 0$. The LD sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) < -RD_{t-1}$. $EBTRD_t$ is earnings before taxes and R&D.

[b] The Cut (Increase) sample consists of all firm-years for which the change in R&D per share is negative (nonnegative).

(3) fewer than three other firms are available in the same industry (4-digit SIC) to calculate industry R&D intensity, (4) data needed to calculate independent variables are missing or equal to zero (which prevents the calculation of log differences in year-to-year amounts), and (5) institutional ownership data is not available on Spectrum. The final sample consists of 13,994 firm-years.

Panel B of table 2 summarizes the number of observations in the SD, LD and IN samples, and the number of firm-years for which R&D was cut in each case.[17] The table shows a significant percentage of firm-year observations with R&D cuts in each sample, and that the number of firm-year observations with R&D cuts in the SD and LD samples are roughly equivalent. It also shows that a majority of firm-year observations in the SD sample do not cut R&D and, hence, do not boost earnings through R&D cuts.[18] These findings indicate that managerial myopia incentives do not completely explain differences in R&D investment behavior across the samples. This observation motivates the inclusion of the controls for opportunity set differences, other earnings management proxies, and institutional ownership variables to explain cross-sectional differences in the decision to cut R&D.

Panel A of table 3 presents means and standard deviations for the variables used in the empirical tests for the SD, IN and LD samples. The percentage change in R&D (CRD) is significantly less positive in the SD sample than in either of the other two samples, consistent with the results of Baber et al. (1991). For most of the other variables, the means in the SD sample lie between the means in the other two samples, as might be expected given the differences in earnings performance across samples.

Panel B of table 3 compares the means of these variables for firm-year observations with R&D cuts and R&D increases in the SD sample. These comparisons provide univariate evidence on myopic investment behavior. The significant differences for PCRD, CIRD, TOBQ, CCAPX and CSALES indicate that reduced opportunity sets significantly explain reductions in R&D. Firms that cut R&D in the SD sample are smaller and more highly leveraged, as expected based on the arguments in the prior section. The percentage of institutional holdings is significantly lower for firms in the SD sample that cut R&D, indicating that higher levels of institutional ownership make managers less likely to use R&D cuts to manage earnings. However, the correlation between PIH and SIZE is high ($r = 0.57$); thus, the relation between institutional holdings and the decision to cut R&D must be examined conditional on SIZE.[19]

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[17] I examined the sample by industry and by year to check for any significant clustering of observations. The majority of the sample firms are in manufacturing, pharmaceuticals/chemicals, and software development industries. The proportion of firms in each industry that cut R&D when in the SD sample does not deviate greatly from the proportion of total observations in the industry. The percentage of firms cutting R&D by year ranges from 23 percent (in 1984) to 41 percent (in 1986 and 1987). Other than these three years, there does not appear to be significant time clustering in R&D reductions.

[18] Because the sample design ignores other methods of managing earnings (e.g., accrual manipulation), these findings likely understate the amount of earnings management that occurs in firms with small declines in earnings. Some firms with expected declines in earnings might have managed earnings using other, less costly means to get into the IN sample. Thus, the SD sample is potentially biased toward firms that lack the discretion in their reporting or economic decisions to boost earnings, thereby reducing the power of the tests.

[19] Other than this correlation, the only pairwise correlations larger than 0.20 between independent variables are the correlations between TOBQ and FCF ($r = -.34$), CSALES and CCAPX ($r = 0.27$), and TOBQ and PCRD ($r = 0.24$). Thus, the independent variables appear to be proxying for different effects. To assess the potential impact of more complex forms of multicollinearity, I examined variance inflation factors (VIF) for each variable. VIFs are based on the R^2 from regressing each independent variable on all other independent variables. A VIF of 1 indicates no multicollinearity and a VIF in excess of 10 indicates harmful multicollinearity (Kennedy 1992, 183). All of the VIFs were less than 1.7, indicating that multicollinearity is not a problem in this sample.

V. RESULTS FOR INSTITUTIONAL OWNERSHIP AS A WHOLE

Main Results

Table 4 presents results for the logit regression of the indicator for cuts in R&D (CUTRD) on the control variables and the percentage of institutional ownership (PIH) (see equation (1)). The first column presents coefficients and p-values for the SD sample. Consistent with the univariate tests, most of the opportunity set variables (PCRD, CIRD,

TABLE 3
Descriptive Statistics

Panel A: Descriptive Statistics for Small Decrease (SD), Increase (IN), and Large Decrease (LD) Samples[a]

		Sample			Differences in Means	
Variable		*SD*	*IN*	*LD*	*(SD)–(IN)*	*(SD)–(LD)*
CUTRD[b]	Mean	0.345	0.257	0.352	0.088**	−0.007
	Std. dev.	*0.475*	*0.437*	*0.478*		
CRD	Mean	0.049	0.146	0.121	−0.097**	−0.072**
	Std. dev.	*0.330*	*0.383*	*0.469*		
PCRD	Mean	0.167	0.150	0.157	0.017*	0.010
	Std. dev.	*0.349*	*0.418*	*0.506*		
CIRD	Mean	0.110	0.110	0.091	0.000	0.019*
	Std. dev.	*0.257*	*0.250*	*0.283*		
CGDP	Mean	0.027	0.030	0.027	−0.003**	0.000
	Std. dev.	*0.017*	*0.017*	*0.018*		
TOBQ	Mean	1.536	1.856	1.378	−0.320**	0.158**
	Std. dev.	*1.741*	*1.896*	*1.593*		
CCAPX	Mean	−0.019	0.106	−0.108	−0.125**	0.089**
	Std. dev.	*0.784*	*0.824*	*0.895*		
CSALES	Mean	0.027	0.181	−0.068	−0.154**	0.095**
	Std. dev.	*0.243*	*0.271*	*0.315*		
SIZE	Mean	4.300	4.725	3.724	−0.425**	0.576**
	Std. dev.	*2.173*	*2.190*	*2.041*		
DIST	Mean	−0.417	2.279	−4.949	−2.696**	4.532**
	Std. dev.	*0.283*	*9.335*	*12.921*		
LEV	Mean	0.184	0.180	0.251	0.004	−0.067**
	Std. dev.	*0.178*	*0.172*	*0.227*		
FCF	Mean	−0.156	−0.054	−0.257	−0.102**	0.101**
	Std. dev.	*0.417*	*0.453*	*0.525*		
PIH	Mean	0.233	0.264	0.190	−0.031**	0.043**
	Std. dev.	*0.208*	*0.228*	*0.192*		
N		2469	8520	2955		

(*Continued on next page*)

TABLE 3 (Continued)

Panel B: Descriptive Statistics for SD Sample by Cut or Increase in R&D

Variable		*Sample*[c] *Cut*	*Sample*[c] *Increase*	*Differences in Means* *(Cut)–(Increase)*	*Expected Sign*[d]
CRD	Mean	−0.248	0.206	−0.454**	
	Std. dev.	*0.314*	*0.208*		
PCRD	Mean	0.124	0.190	−0.066**	−
	Std. dev.	*0.367*	*0.337*		
CIRD	Mean	0.079	0.127	−0.048**	−
	Std. dev.	*0.256*	*0.256*		
CGDP	Mean	0.026	0.027	−0.001	−
	Std. dev.	*0.016*	*0.017*		
TOBQ	Mean	1.420	1.598	−0.178**	−
	Std. dev.	*1.759*	*1.729*		
CCAPX	Mean	−0.225	0.089	−0.314**	−
	Std. dev.	*0.794*	*0.756*		
CSALES	Mean	−0.035	0.060	−0.095**	−
	Std. dev.	*0.249*	*0.233*		
SIZE	Mean	3.938	4.491	−0.557**	−
	Std. dev.	*2.242*	*2.111*		
DIST	Mean	−0.424	−0.414	−0.010	+
	Std. dev.	*0.281*	*0.284*		
LEV	Mean	0.202	0.174	0.028**	+
	Std. dev.	*0.205*	*0.161*		
FCF	Mean	−0.156	−0.156	0.000	−
	Std. dev.	*0.414*	*0.418*		
PIH	Mean	0.201	0.250	−0.049**	+/−
	Std. dev.	*0.198*	*0.212*		
N		852	1617		

* Significantly different from zero (two-tailed) at or below the 0.05 level.
** Significantly different from zero (two-tailed) at or below the 0.01 level.
[a] The SD sample consists of all firm-years for which $-RD_{t-1} < (EBTRD_t - EBTRD_{t-1}) < 0$. The IN sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) > 0$. The LD sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) < -RD_{t-1}$. $EBTRD_t$ is earnings before taxes and R&D.
[b] CRD is the log change in R&D per share. All other variables are defined in table 1.
[c] The Cut (Increase) sample consists of all firm-years for which the change in R&D per share is negative (nonnegative).
[d] This column lists each variable's expected relation to CUTRD.

CCAPX, CSALES) are significantly associated with the probability of cutting R&D in the direction anticipated. The coefficient on SIZE indicates that smaller firms are significantly more likely to cut R&D, consistent with the explanations that small firms are more cash

TABLE 4
Logit Regression of Indicator for Cut in R&D (CUTRD) on Control Variables and Institutional Holdings

$$Prob(CUTRD_i = 1) = F(\alpha + \beta_1 PCRD_i + \beta_2 CIRD_i + \beta_3 CGDP_i + \beta_4 TOBQ_i + \beta_5 CCAPX_i + \beta_6 CSALES_i + \beta_7 SIZE_i + \beta_8 DIST_i + \beta_9 LEV_i + \beta_{10} FCF_i + \beta_{11} PIH_i)$$

		SD Sample[a]		*IN Sample*		*LD Sample*	
Variable	*Expected Sign*	*Coefficient (p-value)*	*Marginal Effects*[c]	*Coefficient (p-value)*	*Marginal Effects*	*Coefficient (p-value)*	*Marginal Effects*
Intercept[b]		−0.158 *(0.323)*		0.372 *(0.000)*		0.071 *(0.596)*	
PCRD	−	−0.448 *(0.001)*	−0.029	−0.335 *(0.000)*	−0.018	−0.249 *(0.005)*	−0.020
CIRD	−	−0.677 *(0.000)*	−0.027	−0.706 *(0.000)*	−0.020	−0.600 *(0.000)*	−0.024
CGDP	−	0.399 *(0.882)*	0.001	−3.398 *(0.036)*	−0.005	−6.138 *(0.008)*	−0.020
TOBQ	−	−0.020 *(0.497)*	−0.004	0.018 *(0.300)*	0.004	−0.066 *(0.044)*	−0.001
CCAPX	−	−0.443 *(0.000)*	−0.075	−0.404 *(0.000)*	−0.054	−0.219 *(0.000)*	−0.044
CSALES	−	−1.464 *(0.000)*	−0.056	−2.161 *(0.000)*	−0.082	−1.526 *(0.000)*	−0.087
SIZE	−	−0.054 *(0.031)*	−0.030	−0.226 *(0.000)*	−0.117	−0.055 *(0.033)*	−0.031
DIST	+[d]	0.072 *(0.649)*	0.007	−0.036 *(0.000)*	−0.012	0.061 *(0.000)*	0.043
LEV	+	0.782 *(0.002)*	0.041	0.769 *(0.000)*	0.030	0.613 *(0.002)*	0.038
FCF	−	0.064 *(0.595)*	0.004	−0.349 *(0.000)*	−0.015	−0.076 *(0.479)*	−0.005
PIH	+/−	−0.928 *(0.001)*	−0.069	−0.030 *(0.859)*	−0.002	−0.412 *(0.144)*	−0.026
pseudo-R^2		0.122		0.220		0.121	
N		2469		8520		2955	

[a] The SD sample consists of all firm-years for which $-RD_{t-1} < (EBTRD_t - EBTRD_{t-1}) < 0$. The IN sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) > 0$. The LD sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) < -RD_{t-1}$. $EBRD_t$ is earnings before taxes and R&D.
[b] All variables are defined in table 1.
[c] The marginal effect represents the change in the probability that the firm cuts in R&D given a change in the independent variable over the interquartile range (i.e., from the 25th percentile to the 75th percentile).
[d] The expected sign for DIST only applies to the SD sample. If earnings smoothing (big bath) incentives motivate managers of firms in the IN (LD) sample, a positive (negative) sign is expected in the IN (LD) sample.

constrained and/or have greater information asymmetries which allow them to use R&D to manage earnings. The coefficient on LEV is significantly positive, consistent with earnings management based on debt covenant incentives and/or reduced growth opportunities. Neither free cash flow (FCF) nor the distance from the earnings goal (DIST) variable is significant.

Turning to the main result, the coefficient on the PIH variable is significantly negative. This result supports H1b, suggesting that institutional investors are sophisticated investors who serve a monitoring role, relative to individual investors, in reducing the incentives of managers to reverse a decline in earnings through cuts in R&D. This finding does not support the concern, expressed in H1a, that institutional investors cause underinvestment in R&D based on their expected reactions to disappointing current earnings news.

The second column of table 4 presents the estimated impact of a change in each independent variable across its interquartile range on the probability of cutting R&D. To obtain this impact, I multiply the marginal effect of a one-unit change in the independent variable (which is calculated as the estimated coefficient times the logit density function evaluated at the sample means of the independent variables) by the interquartile range for each variable. A change in institutional holdings across the interquartile range (from 5 percent to 38 prcent) reduces the probability that a manager cuts R&D by 6.9 percent. This effect appears economically significant, given that about 30 percent of all firms cut R&D in any given year. Only the CCAPX variable has a larger estimated impact in this sample than the PIH variable.

A possible explanation for the negative relation between institutional ownership and the decision to cut R&D is that institutions prefer to invest in more innovative firms that are unlikely to cut R&D in any circumstance. If this were the case, the negative relation should persist outside of the SD sample. Columns 3–6 of table 4 present results for the logit model in the IN and LD samples. The percentage of institutional holdings has no significant influence on the decision to cut R&D out of the SD sample. These results suggest that institutional ownership exerts a greater influence on R&D investment decisions when managers face a trade-off between maintaining R&D and increasing earnings.

Columns 3–6 of table 4 also indicate that the coefficients on the opportunity set controls retain their sign and significance in the IN and LD samples, indicating that these variables explain R&D changes equally well across firm-years with different earnings performance. The coefficients on the SIZE and LEV variables also maintain their significance regardless of the realization of current year earnings. Thus, these variables are likely capturing opportunity set determinants of R&D spending rather than proxying for earnings management motivations. Finally, the coefficient on the DIST variable is significant in the IN and LD samples. In the IN sample, the negative coefficient on DIST indicates that firm-years with the largest (smallest) positive changes in earnings relative to prior R&D expense are less (more) likely to cut R&D, which is consistent with the best performers dampening their earnings growth through increased R&D spending. In the LD sample, the positive coefficient on DIST suggests that extremely poor performers are less likely to cut R&D. This indicates that these firms either take a "big bath" with increased R&D spending because they are far below their earnings target or that these are firms in an early high-investment-low-earnings stage of their life cycle.

Robustness Tests

Another possible explanation for the negative relation between institutional ownership and the decision to cut R&D in the SD sample is that, based on past appearances by firms in this sample, institutional investors invest in firms that are less likely to cut R&D when faced with a small decline in earnings. To test this possibility, I examine whether firms that made repeat occurrences in the SD sample drive the results in table 4. About 50 percent of the firm-year observations in the SD sample are also in the sample in another year, and an additional 25 percent are in the SD sample an additional two years. I partition the sample into firms that made multiple appearances in the SD sample and those that made only a single appearance, and run the regression in equation (1) in each partition. The coefficient on PIH is negative and significant at the 0.05 level in both partitions (not reported). Thus, the results in table 4 are not driven by firms that repeatedly fall in the SD sample.

Because institutional ownership is positively associated with the level of analyst following (O'Brien and Bhushan 1990), another possible explanation for the results in table 4 is that managers are less likely to cut R&D given a large analyst following. Analyst following proxies for a richer information environment for the firm and less uncertainty about the quality of investment spending. As an additional control variable, I added the number of analysts following the firm, defined as the number of analysts issuing a forecast of annual earnings on the I/B/E/S database at the time that institutional holdings are collected.[20] The coefficient on this variable was insignificant, and the sign and significance of the PIH coefficient was unaffected by the inclusion of this variable.

I tested whether the results are sensitive to the definition of the SD sample, specifically the assumptions that prior year's earnings is the target and that prior year's R&D is the maximum amount of R&D that could be cut to meet the target. I modified the tests to (1) specify the lower bound of the SD sample to be pre-R&D earnings changes equal to one-half, one-third, one-fourth, one-fifth, and one-tenth of prior year's R&D, (2) define the earnings target using a random walk with drift model, and (3) use the Hunt et al. (1996) time trend algorithm for determining the earnings target. In each case, the coefficient of the PIH variable is significant at the 0.01 level in the SD sample and insignificant in the other samples.[21]

I added additional variables to control for potential differences in the costs of cutting R&D across firms. Some R&D projects, such as basic research, are easier for managers to justify dropping because of the lack of immediate benefits and highly uncertain future payoffs (Schneiderman 1991). Since no public data are available regarding the composition of R&D expenditures of individual firms, I use proxies for differences in the average composition of R&D across industries based on survey results from Mansfield (1981). I include indicator variables for firms in industries in which R&D has a higher component of (1) basic research, (2) long-term projects, (3) new projects, and (4) risky projects. As an alternative approach, I add an indicator variable for firms in "high-tech" industries, as defined by Chan et al. (1990). They find that the market reacts positively to R&D increases for high-tech firms, but not for "low-tech" firms, indicating that firms in low-tech industries generally have less valuable R&D and would be more likely to cut R&D to meet earnings goals. None of the additional control variables are significant, likely because they are poor proxies for the costs of cutting R&D for an individual firm. The inclusion of these proxies does not affect the significance of the PIH variable.

I also added additional variables to control for other differences in ability or incentives to maintain earnings growth. I reran the tests adding an indicator variable for firms with at least five years of increasing earnings. Barth et al. (1997) find that such firms have strong

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[20] I would like to thank I/B/E/S International Inc. for providing this data.

[21] I also tested the results with the SD sample based on quarterly earnings performance and quarterly R&D expenses for a subsample of firms voluntarily reporting quarterly R&D on Compustat. In this specification, the coefficient on the PIH variable in the SD sample was negative, but not significant. The lack of significant results likely stems from lower power due to a smaller sample (the sample for these tests have less than 400 observations) and to the fact that firms voluntarily reporting quarterly R&D have significantly higher levels of institutional ownership than firms not reporting quarterly R&D, reducing the dispersion in the PIH variable.

incentives to maintain their earnings growth to avoid a significant drop in their pricing multiple. I also included future values of the change in R&D, change in CAPX, change in sales, and change in pre-R&D earnings to proxy for ability/incentives to smooth earnings. The only variable with a significant coefficient in the SD sample was the next year's change in R&D. This variable had a negative coefficient, indicating that firms which cut R&D in the test year had smaller changes in R&D the next year. The inclusion of these variables also did not affect the significance of the PIH variable.

Finally, I tested whether the decision to cut R&D is a function of the change (or expected change) in institutional holdings. I reran the analysis adding the contemporaneous and future changes in institutional holdings as independent variables. I examined future changes up to five years subsequent to the test year. The coefficients on these variables were insignificant in the SD sample, indicating that managers' decisions to cut or increase R&D in this sample do not change the opinions of institutions to the extent that they significantly increase or decrease their holdings.

VI. RESULTS FOR OWNERSHIP BY DIFFERENT GROUPS OF INSTITUTIONS

Classification of Institutional Investors

This section describes how I use factor analysis and cluster analysis to classify institutional investors into groups based on their past investment behavior. Using prior research and guides on institutional ownership, I constructed nine variables that describe the past investment behavior of institutional investors (Baysinger et al. 1991; Porter 1992; Carson Group 1995). I use four variables to proxy for the level of portfolio diversification of each institution (see table 5 for definitions). The level of portfolio concentration (CONC) measures the average percentage of an institution's total equity holdings invested in each portfolio firm. The average percentage holding variable (APH) measures the average size of the institution's ownership position in its portfolio firms. Two variables capture how much of the institution's total equity is invested in firms for which the institution has an influential ownership position. LBPH is the percentage of the institution's equity that is invested in firms where it has greater than 5 percent ownership and HERF is a Herfindahl measure of concentration calculated using the squared percentage ownership in each portfolio firm.

I include two variables to measure the degree of portfolio turnover of the institution. Portfolio turnover (PT) measures the average absolute change in the institution's ownership positions over a quarter, scaled by the change in total equity of the institution. The relative stability of the institution's holdings in its portfolio firms (STAB) measures the percentage of the institution's total equity invested in firms that it has continuously held for the prior two years.

Finally, I include three variables to measure the institution's trading sensitivity to current earnings. The first variable, CETS1, measures the interaction between changes in an institution's holdings in a firm over a quarter and that firm's seasonal change in quarterly earnings announced during the quarter. The value of CETS1 will be high (low) if the institution invests more heavily in firms with larger positive (negative) earnings changes and divests more heavily of firms with negative (positive) earnings. Since this measure assumes a linear relation between the magnitude of the change in holdings and the earnings change, I use two additional measures that allow for nonlinear relations. The second variable, CETS2, measures the difference between the average earnings change of firms in which the institution increased its holdings and the average earnings change of firms in

TABLE 5
Institutional Investor Characteristics[a]

Variable	*Definition*
Portfolio concentration (CONC)	$\Sigma\ w_{kt}/NSTK_t$
Average percentage holding (APH)	$(\Sigma\ w_{kt}\ PH_{kt})/\Sigma\ w_{kt}$
Percent held in large blocks (LBPH)	$(\Sigma\ w_{kt}\ LB_{kt})/\Sigma\ w_{kt}$
Herfindahl measure of concentration (HERF)	$\ln\ (\Sigma\ PH_{kt}^2)$
Stability of holdings (percent held for two years) (STAB)	$(\Sigma\ w_{kt}\ LT_{kt})/\Sigma\ w_{kt}$
Portfolio turnover (PT)	$\Sigma\lvert\Delta\ w_{kt}\rvert/(\Sigma\ w_{kt} + \Sigma\ w_{k,t-1})$
Trading sensitivity to current earnings (CETS1)	$(\Sigma\Delta\ w_{kt}\ RWE_{kt})/\Sigma\lvert\Delta\ w_{kt}\rvert$
Average earnings change of firms bought vs. firms sold (CETS2)	Avg. $(RWE_{kt}\lvert\Delta\ w_{kt} > 0)$–Avg. $(RWE_{kt}\lvert\Delta\ w_{kt} < 0)$
Change in holdings in firms with positive earnings vs. firms with negative earnings (CETS3)	$[(\Sigma\Delta\ w_{kt}\lvert RWE_{kt} > 0)-(\Sigma\Delta\ w_{kt}\lvert RWE_{kt} < 0)]/\Sigma\lvert\Delta\ w_{kt}\rvert$

$NSTK_t$ = number of stocks owned by institution at end of quarter t;
w_{kt} = portfolio weight (shares held times stock price) in firm k at end of quarter t;
$\Delta w_{kt} = w_{kt} - w_{kt-1}$;
PH_{kt} = percentage of total shares in firm k held by institution at end of quarter t;
LB_{kt} = 1 if $PH_{kt} > 0.05$, 0 otherwise;
LT_{kt} = 1 if institution held firm k continuously for prior eight quarters, 0 otherwise;
RWE_{kt} = seasonal random walk change in quarterly earnings per share of firm k for quarter t (deflated by sales for quarter t–4);

[a] The characteristics are calculated at the end of each calendar quarter for every institution on the Spectrum database. The quarterly values are averaged over all quarters available for the calendar year to get end-of-year average values of each characteristic for each institution. These average annual values are used in the subsequent factor and cluster analyses.

which the institution reduced its holdings. The third variable, CETS3, measures the difference between the total change in holdings of firms with positive earnings changes and the total change in holdings of firms with negative earnings changes.

Many of these portfolio characteristics are highly correlated with each other, making it difficult to draw conclusions based on a single characteristic or to include multiple characteristics in the same analysis. To mitigate this problem, I perform principal factor analysis with an oblique rotation to identify and interpret common factors.[22] This procedure reduces

[22] The principal factor approach is appropriate when the goal of the analysis is exploratory rather than confirmatory, and is adopted in a similar context by Dechow et al. (1996). I chose an oblique rotation over an orthogonal (varimax) rotation because it does not rely on the assumption that the factors are independent. However, a varimax rotation yields virtually identical results.

the dimensionality of the data by combining the nine individual characteristics into a parsimonious set of linear combinations that explain the shared variance among the characteristics. Once common factors are identified, I calculate standardized factor scores for use in the subsequent cluster analysis.

Panel A of table 6 presents the results of the factor analysis, which produced three common factors.[23] The BLOCK factor captures the average size of an institution's stake in its portfolio firms. Institutions with high (low) BLOCK scores have portfolios characterized by larger (smaller) average investments in their portfolio firms. The PTURN factor measures the degree of portfolio turnover. Institutions with high (low) PTURN scores trade more (less) frequently and are less (more) likely to hold any given firm in their portfolio continuously for two years. The MOMEN factor captures the trading sensitivity to current earnings news. Institutions with high MOMEN scores are momentum traders who tend to increase (decrease) their holdings in stocks with positive (negative) current earnings news; whereas institutions with low MOMEN scores are contrarian traders who tend to increase (decrease) their holdings in stocks with negative (positive) current earnings news.

I perform k-means cluster analysis on the factor scores to obtain the final separation of institutions into groups. Cluster analysis is a technique for grouping observations into clusters so that institutions are more similar to institutions in the same cluster than they are to institutions in other clusters.[24] There are no standard objective criteria for choosing the number of clusters. Hair et al. (1995) suggest computing a number of different cluster solutions, within a reasonable prespecified range, and deciding among the alternatives based on *a priori* criteria, theoretical considerations, or practical judgment. My goal was to find cluster solutions without disproportionate numbers of observations in some clusters (e.g., one cluster with 90 percent of the observations and the rest with the remaining 10 percent) that closely match the theoretical reasons for splitting institutions into groups.

Panel B of table 6 shows the results of the cluster analysis. The three-cluster solution yields groups matching the definitions of transient, dedicated, and quasi-indexer institutions. Based on the mean factor scores for each cluster, "transient" institutions have the highest turnover (PTURN) and highest use of momentum strategies (MOMEN), along with relatively high diversification (small BLOCK). "Dedicated" institutions have high concentration, low turnover, and almost no trading sensitivity to current earnings (average MOMEN near zero). "Quasi-indexers" exhibit high diversification and low turnover, consistent with index-type, buy-and-hold behavior. Quasi-indexers also exhibit contrarian-trading tendencies (low MOMEN), which are consistent with most buy-and-hold value strategies.

The quasi-indexers are the largest class of institutional investors, accounting for 70 percent of the sample. Since only about 10 percent of institutions have stated index strategies (Carson Group 1995), the majority of the quasi-indexer group does not explicitly index, but follow longer-term buy-and-hold strategies. The predominance of this group is likely due to (1) the fact that the transient group captures only the most aggressive momentum traders, which comprise a relatively small portion of institutional strategies (Carson Group 1995); and (2) the aggregation of all individual funds within a fund group (e.g., all

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[23] I used three criteria to determine the appropriate number of factors to keep: scree plots, minimum eigenvalue, and proportion of variance explained. All three criteria produced consistent results.

[24] The k-means method of cluster analysis starts with a set of cluster seeds and then assigns observations into clusters based on the Euclidean distances between the observation and the means of the cluster seeds (which change as additional observations are added). With large numbers of observations, the formation of clusters using this method is insensitive to the selection of initial seeds and to the order of the observations (Hair et al. 1995).

TABLE 6
Factor Analysis and Cluster Analysis of Institutional Investor Portfolio Characteristics

Panel A: Factor Analysis[a]

Variable	*Factors* *BLOCK*	*PTURN*	*MOMEN*
APH[b]	0.818	0.019	−0.013
LBPH	0.798	−0.023	−0.004
HERF	0.338	0.063	0.073
CONC	0.327	−0.062	−0.009
PT	0.033	0.766	−0.003
STAB	0.036	−0.751	0.015
CETS1	0.022	0.004	0.727
CETS2	0.004	−0.009	0.722
CETS3	0.009	−0.015	0.053
Variance Explained	43.2%	30.7%	26.1%

Panel B: Clusters Based on Factor Scores[c]

Cluster	*N*[e]	*Mean Factor Scores*[d] *BLOCK*	*PTURN*	*MOMEN*
Quasi-Indexers	6060	−0.153	−0.250	−0.272
Transient	2225	−0.075	0.337	0.688
Dedicated	319	3.303	−0.625	−0.078

Panel C: Persistence of Cluster Membership Across Time

	Percent in Cluster—Year t+1			*Percent in Cluster—Year t+3*		
Year t Cluster	*Quasi-Indexers*[f]	*Transient*	*Dedicated*	*Quasi-Indexers*	*Transient*	*Dedicated*
Quasi-Indexers	80.2	19.2	0.6	78.5	20.5	1.0
Transient	56.7	42.5	0.8	57.5	41.2	1.3
Dedicated	10.2	4.1	85.8	18.9	4.9	76.2

[a] Factors are estimated using principal factor analysis with an oblique promax rotation.
[b] All variables are defined in table 5.
[c] Clusters are formed using k-means cluster analysis.
[d] The factor scores are generated from the factors in panel A.
[e] N represents the number of institution-years in each cluster.
[f] The percentages in each column represent the percentage of institutions in each year t cluster that are in each year t+1 (or t+3) cluster, excluding institutions dropping out of the sample prior to year t+1 (t+3).

of Fidelity's funds are listed in one holdings figure).[25] The low number of dedicated institutions (4 percent) is not surprising given that institutions often face internal restrictions that limit the size of any stake they can take in an individual firm, and have fiduciary and governance incentives that make diversification a more desirable strategy (Monks and Minow 1995, chap. 2). Overall, the relative numbers of institutions in each group support Porter's (1992, 8) argument that U.S. institutional investors are unlikely to take substantial ownership positions in firms, but rather are likely to take small, highly diversified stakes and/or turnover these positions frequently.

Panel C of table 6 documents the persistence of cluster membership across time. Institutions classified as quasi-indexer and dedicated tend to maintain that classification into the future, as over 80 percent (75 percent) of these institutions are in the same cluster group one (three) years hence. On the other hand, the majority of transient institutions move to another group in subsequent periods. In the next year (and three years hence), only about 40 percent of transient institutions remain classified as such; over 55 percent are classified in the quasi-indexer category in subsequent years. The factor that drives the instability of the transient classification is the MOMEN factor, which is higher in the year the institution is classified as transient. Thus, many institutions appear to change the amount of momentum trading they do over time, indicating that such institutional trading strategies are not as persistent across time as might be expected.

Results for Predominant Ownership by Different Groups of Institutional Investors

Table 7 presents results for the logit regression of CUTRD on the control variables and institutional holdings including the indicator variables for predominant ownership by transient, dedicated, and quasi-indexer institutions (see equation (2)). The restriction that firms have at least 5 percent institutional ownership reduces the sample sizes relative to table 4 (see the last row of table 7). Column 1 presents results for the SD sample. The only change in the significance of the control variables between table 4 and table 7 is that the coefficient on SIZE is no longer significant. This lack of significance is caused by constraining the sample to only include firms with greater than 5 percent institutional ownership (which are primarily larger firms) and not due to the addition of the indicator variables for holdings by group. Consistent with H2a, firms with an extremely high proportion of ownership by transient institutions are significantly more likely to cut R&D.[26] The coefficient on the transient indicator variable is insignificant in both the IN and LD samples, indicating that extreme transient ownership only influences the decision to cut R&D for firms with small declines in earnings. This evidence supports the concern of Porter (1992), Jacobs (1991), Graves and Waddock (1990), and others that transient institutional investor behavior leads to myopic R&D investment behavior.

The results indicate that extreme proportions of ownership by dedicated institutions and by quasi-indexers have no incremental impact on the likelihood of R&D cuts in any of the subsamples. For dedicated institutions, the lack of any significant incremental impact probably stems from the fact that any effect of dedicated ownership is difficult to detect due to the limited number of cases in which dedicated institutions own majority shares of

[25] Many institutions that maintain numerous funds, such as Fidelity and Dreyfus, are split between transient and quasi-indexer categories across years in the sample. These switches in category likely occur when one set of funds within the fund family has an exceptionally good year or receives a larger influx of new money, tilting the average portfolio characteristics toward these funds.

[26] The coefficient on the transient indicator variable is also significantly positive ($p = 0.03$) when defined using quartiles, and insignificantly positive ($p = 0.16$) when defined using treciles.

TABLE 7
Logit Regression of Indicator for Cut in R&D (CUTRD) on Control Variables and Institutional Holdings By Group of Institution

$$Prob(CUTRD_i = 1) = F[\alpha + \beta_1 PCRD_i + \beta_2 CIRD_i + \beta_3 CGDP_i + \beta_4 TOBQ_i + \beta_5 CCAPX_i + \beta_6 CSALES_i + \beta_7 SIZE_i + \beta_8 DIST_i + \beta_9 LEV_i + \beta_{10} FCF_i + \beta_{11} PIH_i + \beta_{12} DQ5(Transient)_i + \beta_{13} DQ5(Quasi\text{-}indexer)_i + \beta_{14} DQ5(Dedicated)_i]$$

		SD Sample[a]		*IN Sample*		*LD Sample*	
Variable	*Expected Sign*	*Coefficient (p-value)*	*Marginal Effects*[c]	*Coefficient (p-value)*	*Marginal Effects*	*Coefficient (p-value)*	*Marginal Effects*
Intercept[b]		−0.184 *(0.424)*		0.595 *(0.000)*		0.207 *(0.305)*	
PCRD	−	−0.484 *(0.008)*	−0.029	−0.629 *(0.000)*	−0.027	−0.379 *(0.004)*	−0.027
CIRD	−	−0.982 *(0.000)*	−0.037	−0.742 *(0.000)*	−0.019	−0.762 *(0.000)*	−0.030
CGDP	−	−0.791 *(0.820)*	−0.002	−4.877 *(0.025)*	−0.011	−3.332 *(0.286)*	−0.010
TOBQ	−	−0.099 *(0.068)*	−0.016	−0.037 *(0.202)*	−0.007	−0.065 *(0.256)*	−0.008
CCAPX	−	−0.521 *(0.000)*	−0.077	−0.425 *(0.000)*	−0.045	−0.216 *(0.002)*	−0.039
CSALES	−	−1.917 *(0.000)*	−0.062	−3.012 *(0.000)*	−0.092	−1.720 *(0.000)*	−0.084
SIZE	−	−0.031 *(0.390)*	−0.016	−0.212 *(0.000)*	−0.085	−0.071 *(0.045)*	−0.037
DIST	+[d]	0.232 *(0.221)*	0.023	−0.025 *(0.028)*	−0.006	0.060 *(0.000)*	0.037
LEV	+	0.863 *(0.014)*	0.041	0.681 *(0.001)*	0.022	1.058 *(0.000)*	0.060
FCF	−	0.119 *(0.557)*	0.006	−0.461 *(0.000)*	−0.016	0.141 *(0.426)*	0.008
PIH	+/−	−0.946 *(0.016)*	−0.060	−0.096 *(0.671)*	−0.005	−0.269 *(0.655)*	−0.016
DQ5(Transient)	+	0.365 *(0.015)*	0.007	−0.067 *(0.464)*	−0.001	−0.062 *(0.655)*	−0.001
DQ5(Quasi-indexer)	+/−	0.166 *(0.240)*	0.003	−0.144 *(0.116)*	−0.002	0.030 *(0.821)*	0.001

(*Continued on next page*)

TABLE 7 (Continued)

		SD Sample[a]		IN Sample		LD Sample	
Variable	*Expected Sign*	*Coefficient (p-value)*	*Marginal Effects*[c]	*Coefficient (p-value)*	*Marginal Effects*	*Coefficient (p-value)*	*Marginal Effects*
DQ5(Dedicated)	–	0.207 *(0.167)*	0.003	0.107 *(0.200)*	0.002	0.214 *(0.137)*	0.003
pseudo-R^2		0.143		0.221		0.121	
N		1826		6384		1995	

[a] The SD sample consists of all firm-years for which $-RD_{t-1} < (EBTRD_t - EBTRD_{t-1}) < 0$. The IN sample consists of all firm-years for which $(EBTRD_t - EBTRD_{t-1}) > 0$. The LD sample consists of all firms for which $(EBTRD_t - EBTRD_{t-1}) < -RD_{t-1}$. $EBTRD_t$ is earnings before taxes and R&D. The samples are restricted to firms with greater than 5 percent total institutional holdings.

[b] $DQ5(GROUP_k)_i$ equals one if the proportion of ownership by group k (k = quasi-indexer, transient, dedicated) in firm i is in the top quintile of its distribution. All other variables are defined in table 1.

[c] The marginal effect represents the change in the probability that the firm cuts in R&D given a change in the independent variable over the interquartile range (i.e., from the 25th percentile to the 75th percentile).

[d] The expected sign for DIST only applies to the SD sample. If earnings smoothing (big bath) incentives motivate managers of firms in the IN (LD) sample, a positive (negative) sign is expected in the IN (LD) sample.

firms.[27] For quasi-indexer ownership, the lack of results is probably due to the high correlation between extreme quasi-indexer ownership and overall institutional ownership, of which quasi-indexers are the largest group.

VII. SUMMARY AND CONCLUSIONS

This paper contributes to the ongoing debate about the short-term focus of U.S. managers and investors by providing evidence on the influence of institutional investors on the investment behavior of managers facing incentives to cut R&D to maintain earnings growth. The results indicate that managers are significantly less likely to cut R&D to reverse an earnings decline when institutional ownership is high. This evidence supports the view that, relative to individual investors, the large stockholdings and sophistication of institutional investors allows them to monitor and discipline managers, ensuring that managers choose R&D levels to maximize long-run value rather than to meet short-term earnings goals. However, given a significant level of institutional ownership, a high proportion of ownership by institutions exhibiting transient ownership characteristics (i.e., high portfolio turnover, diversification, and momentum trading) significantly increases the probability that managers reduce R&D to boost earnings. This result supports the widely-argued view that short-term-oriented behavior by institutions creates pressures for managers to sacrifice R&D for the sake of higher current earnings, but only for a sample of firms with extremely high levels of transient ownership and not for the U.S. market, or for institutional investors, as a whole.

Some important caveats must be mentioned in these interpretations of the results. First, the evidence on the relation between transient institutional ownership and the decision to

[27] Within the top quintile of proportional ownership by dedicated institutions, the mean holdings of dedicated institutions is less than 20 percent, compared to mean holdings of 94 percent (70 percent) for quasi-indexer (transient) institutions in the top quintile of their respective distributions. As a result, 21[25] firms have values of 1 for both DQ5(Dedicated) and DQ5 (Quasi-indexer) [DQ5(Transient)] in the SD sample.

cut R&D must be interpreted with some caution due to the time-series instability of membership in the transient group. This instability indicates that transient behavior does not persist over time, but this fact works against a finding that past institutional behavior influences current managerial decisions. Second, the theory and the interpretations of the findings are framed in terms of institutional investor influence on managerial decisions to reduce R&D to manage earnings; but in an association test, it is always difficult to establish causality. I address other explanations for the relation, such as self-selection by institutions in the firms they choose, by examining various sample partitions, by comparing changes in R&D to levels of institutional ownership, and by examining concurrent and future changes in institutional ownership. The body of evidence produced by these approaches is most consistent with institutional investors influencing managerial behavior rather than vice versa. Finally, the process by which managers can manipulate R&D to meet earnings goals is not widely understood, and is difficult to document with publicly available data. This limitation makes it difficult to include specific controls for differences in the cost of cutting R&D across firms, and ultimately makes it difficult to discern whether managers permanently cut R&D or temporarily postpone it across fiscal periods. I try to overcome this limitation by including control variables, such as indicators for the average composition of R&D in the firm's industry and future values of R&D and earnings, but more detailed, small sample research is needed to fully understand the mechanics of managerial myopia.

Given these caveats, this paper provides the first large-sample cross-sectional evidence of institutional investor behavior contributing to the myopic investment problem in the U.S. I find this evidence by focusing on a specific situation—reducing R&D to reverse an earnings decline—and by partitioning institutions directly on the investment behavior argued to create incentives for myopic investment behavior. In doing so, this paper also develops a new method for classifying institutional investors into groups based on past investment behavior. This classification is applicable to other contexts, such as explaining market reactions to earnings announcements, that will be explored in future research. Finally, this study motivates further research into the current trend toward more active investor targeting and investor relations by firms as the evidence indicates that different compositions of institutional owners affect managerial decisions.

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